

OCI DevOps Toolkit IntelliJ Plugin — Technical Specification

Description:

OCI DevOps Toolkit is an IntelliJ IDEA plugin that brings DevOps Portal <https://devops.oci.oraclecorp.com> capabilities into the IDE. It provides a unified tool window for SCM pull requests/review, build/pipeline/deployment visibility. Objective: reduce context switching and speed up developer workflows.

Use Case:

For PR review, build/deployment tracking need to switch IDE & Browser. Having information integrated with IDE can reduce friction. The need is a single in-IDE DevOps experience to review PRs, monitor pipeline/deployment activity. By leveraging [DevOps API](#) and [OCI CLI](#)

Enhancement:

- Fetching SCM pull request , search, sorting, filtering (user/status/branch).
- Build and deployment tabs for run/pipeline/deployment insights (expand scaffolded tabs with real OCI data/actions).
- Footer services for OCI CLI version, session expiry tracking, and auto-refresh/re-authentication signals.
- Non-blocking background operations, robust error notifications, and safe UI-thread updates.

Scope:

In Scope:

- IDE-native OCI DevOps visibility and interaction
- PR operations dashboard behaviors
- Build/deployment observability panels
- Session and CLI health status in footer
- More Feature: TBD

Functional Requirements

ID	Requirement	Acceptance Criteria
FR-01	PR listing + filter/search/sort	Accurate filtered list, refresh support, empty state shown
FR-02	Build/deployment visibility	Data rendered per tab; unavailable data handled gracefully
FR-03	OCI CLI/session health footer	Version/session status visible and periodically refreshed
FR-04	Async operations + notifications	No UI freeze; actionable error/success notifications

Design Flow

```
flowchart LR
  UI[Tool Window Tabs + Footer] --> SVC[Service Layer]
  SVC --> CLI[OCI CLI Executor]
  SVC --> OCI[OCI DevOps APIs]
  SVC --> EVT[Notification/Event Handler]
```

Tech Stack

- Java
- IntelliJ Platform Plugin SDK
- Gradle (Kotlin DSL)
- OCI CLI + OCI DevOps APIs
- JUnit test framework
- Logback logging
- More Feature: TBD

Key NFRs

- **Performance:** non-blocking UI, responsive panel updates
- **Reliability:** timeout/retry and graceful fallback on OCI/API errors
- **Observability:** structured logs for commands and failures
- **More Feature:** TBD

Delivery Criteria

- Functional requirements mapped to tests
- Plugin packaging and compatibility validated
- More Feature: TBD